

Martin Li

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EDUCATION

University of California, Irvine

Irvine, CA

B.S. Mechanical Engineering | Fourth-Year Undergraduate

Relevant Courses: Statics, Dynamics, Materials Science, Electrical Circuits, Engineering Design, MATLAB

SKILLS

Technical Tools: SolidWorks, MATLAB, Python, R, Cisco networking, 3D printing

Mechanical / Fabrication: Mechanical assembly, hand tools, precision fasteners, prototyping, woodcutting, fit-up, design iteration

Shop / Hardware: Bandsaw, sanding machine, laser engraver, PC component installation and troubleshooting

Engineering Strengths: CAD modeling, component and assembly modeling, testing, troubleshooting, teamwork, technical communication

PROJECT EXPERIENCE

Tripoli Level 1 High-Power Rocket - Aerodynamic Design, CAD & Fabrication

Personal Project

- Designed a high-power rocket with aerodynamic and structural considerations for stable flight and safe recovery.
- Created detailed SolidWorks models, then translated the CAD design into 3D-printed components for fabrication and assembly.
- Integrated and assembled the rocket system, completed pre-launch checks, and prepared the vehicle for field testing at a desert launch site.
- Successfully launched the rocket and earned Tripoli Level 1 high-power rocketry certification.

Two-Legged Running Robot - Mechanical Design & Assembly

UC Irvine

- Led an 8-person engineering team to design and build a functional two-legged running robot under project deadlines.
- Modeled components and the full assembly in SolidWorks to support geometry, part fit, and mechanical layout.
- Assembled the robot with hand tools and precision fasteners; corrected fit and assembly issues during testing.
- Manufactured components through 3D printing and woodcutting, then integrated the fabricated parts into the final build.
- Improved robot functionality through repeated fit checks, assembly adjustments, and design iteration after observed test performance.

Woodshop Manufacturing Project - CAD, Fabrication & Finishing

UC Irvine

- Created a SolidWorks model and template, then translated the digital design into a physical manufactured part.
- Used 3D printing, laser engraving, bandsaw cutting, and sanding equipment to fabricate and finish the part.
- Practiced safe and accurate shop workflow from CAD concept through fabrication, surface preparation, and final finishing.

PC Hardware Assembly & Troubleshooting

Independent Project

- Selected compatible motherboard, CPU, RAM, GPU, power-related hardware, and other components by matching technical specifications.
- Installed and assembled computer hardware while maintaining correct component fit, connections, and system compatibility.
- Diagnosed malfunctioning or worn components using systematic troubleshooting to isolate likely failure points.

TECHNICAL & TEAM EXPERIENCE

Cisco Community Networking Project - Hardware Deployment & Team Lead

Antigua

- Deployed and configured 50+ physical Cisco routers and access points, gaining hands-on field experience with technical hardware and network infrastructure.
- Led a 10-person team and helped coordinate 200+ donated routers and projectors, strengthening organization, equipment handling, and cross-team coordination.
- Worked with Cisco management and local stakeholders to support deployment logistics and provide internet access across schools on the island.

Cisco CCNA Server Room Training

Technical Training

- Configured routers, switches, and servers in a physical lab environment and practiced OSPF, EIGRP, SSH, and full-topology troubleshooting.
- Built practical experience tracing connectivity problems, checking device configuration, and correcting network faults across multi-device setups.